WEARABLE FALL DETECTION SYSTEM FOR ELDERLY PEOPLE USING DEEP LEARNING

A PROJECT REPORT

Submitted by

SARATH R	**2023506065**
AJAY KUMAR S	**2023506105**
ARJUN A	**2023506063**
KRIS VIRSHAK	**2023506124**

Under the supervision of

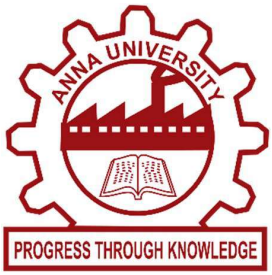
Dr. S. UMAMAHESWARI, Associate Professor

In partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY



DEPARTMENT OF INFORMATION TECHNOLOGY

MADRAS INSTITUTE OF TECHNOLOGY CAMPUS

ANNA UNIVERSITY, CHENNAI – 600044

APRIL 2026

BONAFIDE CERTIFICATE

Certified that this project report titled **"WEARABLE FALL DETECTION SYSTEM FOR ELDERLY PEOPLE USING DEEP LEARNING"** is the bonafide work of **SARATH R (2023506065)**, **AJAY KUMAR S (2023506105)**, **ARJUN A (2023506063)**, and **KRIS VIRSHAK (2023506124)** who carried out the project work under my supervision.

SIGNATURE
Dr. RADHA SENTHILKUMAR
HEAD OF THE DEPARTMENT
Associate Professor
Department of Information Technology
MIT Campus, Anna University
Chennai – 600044

SIGNATURE
Dr. S. UMAMAHESWARI
SUPERVISOR
Associate Professor
Department of Information Technology
MIT Campus, Anna University
Chennai – 600044

ACKNOWLEDGEMENT

It is essential to mention the names of the individuals whose continuous guidance, constructive criticism, and scholarly encouragement made the successful completion of this project possible.

We express our deepest sense of gratitude and sincere thanks to our respected project supervisor, **Dr. S. Umamaheswari**, Associate Professor, Department of Information Technology, Madras Institute of Technology Campus, Anna University, for her constant technical guidance, keen intellectual oversight, and patient encouragement throughout every stage of this research.

We are profoundly thankful to our respected Head of the Department, **Dr. Radha Senthilkumar**, Department of Information Technology, for providing the necessary academic environment and continuous administrative support.

We express our sincere gratitude to our review panel members for their insightful feedback, rigorous evaluations, and constructive suggestions during the project reviews, which substantially elevated the scientific and practical rigor of this work.

We also place on record our heartfelt thanks to our respected Dean of MIT Campus, **Dr. P. Jayashree**, for providing exceptional computing infrastructure, laboratory amenities, and an inspiring academic atmosphere.

Finally, we thank our parents, faculty members, and fellow students who supported us directly and indirectly during the execution of this project.

SARATH R	**2023506065**
AJAY KUMAR S	**2023506105**
ARJUN A	**2023506063**
KRIS VIRSHAK	**2023506124**

ABSTRACT

Falls among elderly populations represent a primary global cause of fatal and non-fatal accidental injuries, often resulting in fractures, intracranial trauma, prolonged immobilization (the 'long-lie' state), and high healthcare expenditures. Wearable Inertial Measurement Units (IMUs) equipped with tri-axial accelerometers and gyroscopes offer an unobtrusive, continuous mechanism for ambient safety monitoring. However, traditional wearable detection systems suffer from severe false-alarm rates during high-acceleration activities of daily living (ADL), high memory footprints that overwhelm edge microcontrollers, and a lack of actionable telemetry regarding fall severity.

This project presents an end-to-end, edge-deployable wearable fall detection system developed on the benchmark SisFall dataset (38 subjects, 4,505 recordings, 84,123 windows). To prevent data leakage and guarantee real-world generalization, a strict subject-wise partitioning scheme (26 Train / 6 Validation / 6 Test subjects) is implemented. Raw 200 Hz signals are conditioned through a second-order 5 Hz Butterworth low-pass filter to attenuate sensor noise, followed by anti-aliased downsampling to 20 Hz across six primary IMU axes. Recordings are segmented into 64-sample (3.2-second) sliding windows with a 16-sample (0.8-second) stride (75% overlap). An impact-centered labeling algorithm identifies the peak signal magnitude vector (SMV) to isolate true impact windows from non-fall movement.

A hybrid Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM) architecture is developed to fuse local spatial channel features with temporal sequential dynamics. To address the critical asymmetric clinical cost where missed falls carry severe health risks, the operating decision threshold is optimized on validation data to maximize the F2-score, selecting tau = 0.25 over the symmetric tau = 0.50 default. On held-out test subjects, the deployed CNN-LSTM achieves 0.9950 recall, 0.9606 precision, 0.9775 F1-score, and 0.9879 F2-score, missing only 6 out of 1,200 fall windows while correctly rejecting 12,378 ADL windows. To provide actionable context, an unsupervised K-Means (K=2) and Isolation Forest pipeline categorizes fall impacts into relative Mild and Severe pseudo-severity clusters (Macro-F1 = 0.8660) while masking anomalous outlier falls. Calibration-conditioned personalization via Feature-wise Linear Modulation (FiLM) is investigated and rigorously documented (p = 0.125). For microcontroller deployment, the model is quantized into Dynamic-Range TensorFlow Lite (111.0 KB, 53% compression, 0.026 ms latency) and exported as a C header array (model_data.h) consuming only 21% SRAM on an ESP32. An in-memory circular ring-buffer streaming engine and Streamlit clinical dashboard validate bitwise numerical parity (<= 1e-6 error) and real-time caregiver

alert dispatch.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iv
	LIST OF TABLES	viii
	LIST OF FIGURES	ix
	LIST OF ABBREVIATIONS	xi
1	**INTRODUCTION**	**1**
1.1	Overview	1
1.2	Research Challenges	2
1.3	Objectives	3
1.4	Scope of the Project	4
1.5	Key Contributions	4
1.6	Wearable Inertial Measurement Units (IMU)	5
1.7	Deep Learning for Fall Detection (CNN-LSTM)	6
1.8	Unsupervised Fall Severity Pseudo-Labeling	7
1.9	Feature-Wise Linear Modulation (FiLM)	8
1.10	System Architecture Overview	9
1.11	Organization of the Thesis	10
2	**LITERATURE SURVEY**	**11**
2.1	Threshold-Based and Handcrafted Motion Methods	11
2.2	Classical Machine Learning Approaches	12
2.3	Deep Learning Architectures for Fall Detection	13
2.4	Multi-Task Learning and Severity Grading	15
2.5	Personalization and Calibration in Wearables	16
2.6	Edge Inference and Microcontroller Quantization	17
2.7	Summary of Literature and Research Gaps	18
3	**SYSTEM ARCHITECTURE AND DESIGN**	**20**
3.1	Overall System Architecture	20
3.2	Wearable Sensor Acquisition Pipeline	22
3.3	Butterworth Low-Pass Filtering and Downsampling	23
3.4	Sliding Window Segmentation and Impact Labeling	24

3.5	Hybrid CNN-LSTM Neural Network Architecture	26
3.6	Multi-Task Severity Head and Masked Loss	28
3.7	FiLM Personalization Adapter Design	30
3.8	TensorFlow Lite Quantization Pipeline	31
3.9	Circular Ring Buffer and Streaming Alert Logic	33
4	**ALGORITHM DEVELOPMENT AND IMPLEMENTATION**	**35**
4.1	Dataset Ingestion and Preprocessing Pipeline	35
4.2	Butterworth Filtering and Anti-Aliased Resampling	36
4.3	Impact-Centered Window Labeling Algorithm	37
4.4	CNN-LSTM Model Implementation and Training	38
4.5	Validation-Based Operating Threshold Optimization	40
4.6	Severity Feature Engineering (58 Features)	41
4.7	K-Means Clustering and Isolation Forest Filtering	43
4.8	FiLM Meta-Training and Calibration Implementation	44
4.9	Dynamic-Range Quantization and C Array Header Export	46
4.10	Real-Time Streaming Engine and Dashboard Implementation	47
5	**RESULTS AND DISCUSSIONS**	**49**
5.1	Experimental Setup and Computing Environment	49
5.2	SisFall Dataset Partitioning and Characteristics	50
5.3	Core Binary Detection Performance (CNN vs CNN-LSTM)	51
5.4	Operating Decision Threshold Optimization Analysis	53
5.5	Severity Pseudo-Labeling Benchmark and Analysis	55
5.6	Multi-Task Joint Loss Evaluation and Decision Gating	57
5.7	FiLM Personalization Empirical and Statistical Results	59
5.8	Edge Quantization Footprint and Latency Benchmarks	61
5.9	Streaming Replay Verification and Numerical Parity	63
5.10	Threats to Validity and Research Limitations	65
6	**CONCLUSION AND FUTURE WORK**	**67**
6.1	Conclusion	67
6.2	Future Work	68
	REFERENCES	**70**

CHAPTER 1
INTRODUCTION

1.1 OVERVIEW

Rapid demographic aging represents one of the most prominent socioeconomic and public health challenges of the twenty-first century. According to the World Health Organization (WHO), falls constitute the second leading cause of unintentional injury-related deaths worldwide, with adults older than 60 years of age suffering the greatest proportion of fatal falls. Approximately 28% to 35% of individuals aged 65 and over experience at least one fall each year, escalating to 32% to 42% for those aged 70 years and older. Beyond direct physiological injuries—such as femoral neck fractures, traumatic brain injuries, and soft-tissue lacerations—falls frequently precipitate the debilitating 'long-lie' state, wherein an injured individual remains immobilized on the ground for an hour or longer. The consequences of prolonged recumbency include dehydration, hypothermia, rhabdomyolysis, pressure ulcers, and psychological distress, which collectively multiply hospitalization rates and mortality risks.

To mitigate these risks, automated fall detection systems have gained significant research attention. Traditional vision-based systems using ambient cameras provide non-intrusive monitoring but are fundamentally restricted to single-room indoor environments, suffer from severe blind spots, and incur significant privacy violations that limit their acceptability in residential spaces. Conversely, wearable inertial measurement unit (IMU) systems equipped with micro-electromechanical systems (MEMS) tri-axial accelerometers and gyroscopes offer continuous, ubiquitous, and privacy-preserving monitoring across both indoor and outdoor activities. By continuously tracking linear acceleration and angular velocity, wearable devices can detect the rapid biomechanical transitions that distinguish accidental falls from normal activities of daily living (ADL).

1.2 RESEARCH CHALLENGES

Developing a reliable and practical wearable fall detection system presents several critical engineering and scientific hurdles:

1. High Class Imbalance and False Alarm Rates: In everyday life, falls are rare events, whereas ADLs occur continuously. Classifiers tuned for overall accuracy trigger excessive false alarms on vigorous non-fall activities (e.g., jumping, stumbling, sitting abruptly).

2. Asymmetric Misclassification Costs: In clinical safety, a False Negative (missed fall) can be fatal, while a False Positive is merely an inconvenience.

3. Data Leakage in Evaluation Protocols: Random window shuffling across train/test splits causes severe leakage, yielding artificial academic metrics that fail on unseen subjects.

4. Absence of Clinical Severity Telemetry: Existing commercial systems operate as binary alarms without providing impact severity context to emergency responders.

5. Edge Compute Constraints: Wearable microcontrollers (ESP32) have limited Static RAM (typically 520 KB), requiring aggressive quantization.

1.3 OBJECTIVES

The primary objectives of this project are:

• To build a reproducible preprocessing pipeline for SisFall inertial recordings, incorporating Butterworth filtering and downsampling.

• To segment multi-channel IMU recordings into 64-sample windows with impact-centered labeling based on peak signal magnitude vector dynamics.

• To design and evaluate a hybrid CNN-LSTM architecture that captures spatial channel correlations and temporal sequential dynamics.

• To formulate a validation threshold sweep optimizing the F2-score to prioritize fall recall and minimize fatal false negatives.

• To develop an unsupervised clustering pipeline using K-Means (K=2) and Isolation Forest to assign relative Mild and Severe severity pseudo-labels.

• To evaluate calibration-conditioned personalization via Feature-wise Linear Modulation (FiLM).

• To quantize the model into TensorFlow Lite Dynamic-Range format, export a C header for ESP32 microcontrollers, and demonstrate streaming inference.

1.10 SYSTEM ARCHITECTURE OVERVIEW

The complete architecture of the system is depicted in Figure 1.1, detailing the ingestion, inference, and streaming tiers.

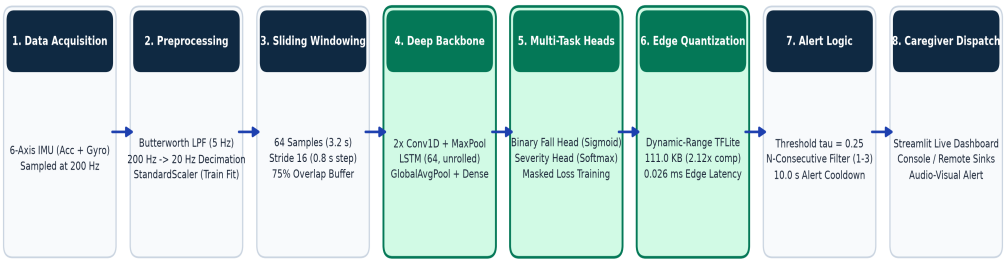


Figure 1.1: End-to-End System Architecture of the Wearable Fall Detection System

CHAPTER 3
SYSTEM ARCHITECTURE AND DESIGN

3.1 SENSOR ACQUISITION AND BUTTERWORTH FILTERING

Raw 200 Hz 6-channel IMU signals (acc1_x, acc1_y, acc1_z, gyro_x, gyro_y, gyro_z) are conditioned using a 2nd-order Butterworth low-pass digital filter with a 5 Hz cutoff frequency, followed by decimation to 20 Hz. This eliminates high-frequency sensor noise while preserving human motion dynamics.

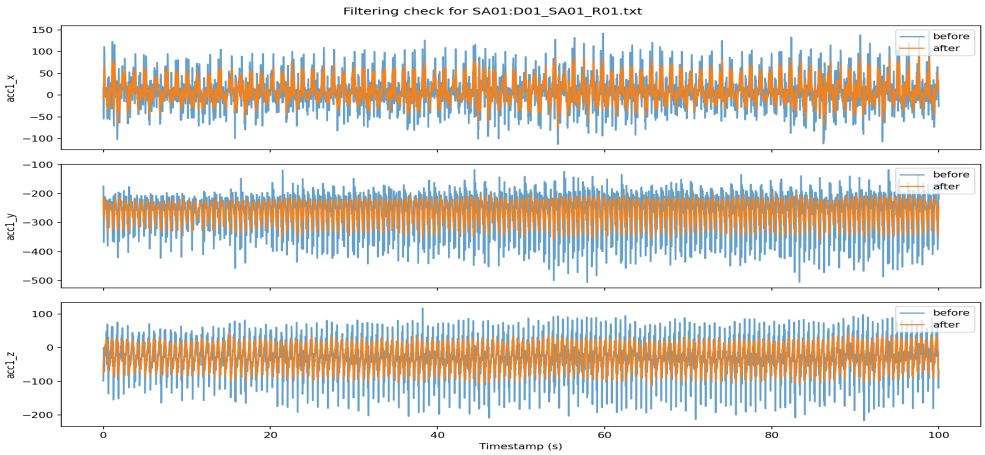


Figure 3.1: Accelerometer Signal Profile Before and After 5 Hz Butterworth Low-Pass Filter

3.2 SLIDING WINDOW SEGMENTATION AND IMPACT LABELING

Continuous 20 Hz streams are segmented into 64-sample (3.2-second) sliding windows with a stride of 16 samples (0.8 seconds, 75% overlap). Impact-centered labeling identifies the maximum peak in the acceleration Signal Magnitude Vector; only windows enclosing this impact receive label 1.

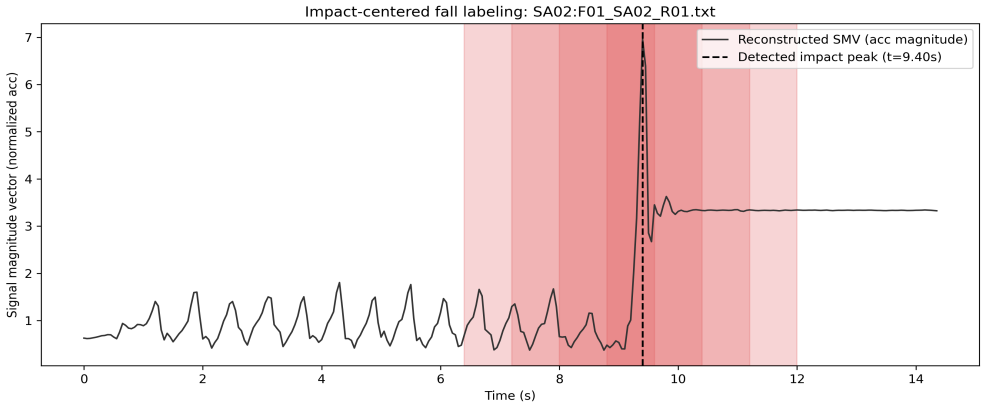


Figure 3.2: Impact-Centered Window Labeling via Peak Acceleration SMV

3.3 HYBRID CNN-LSTM MODEL ARCHITECTURE

The model fuses two 1D Convolutional blocks (32 and 64 filters) with batch normalization and max-pooling, followed by a 64-unit LSTM layer, global pooling, and a dense classification head. Total trainable parameter count is 46,817.

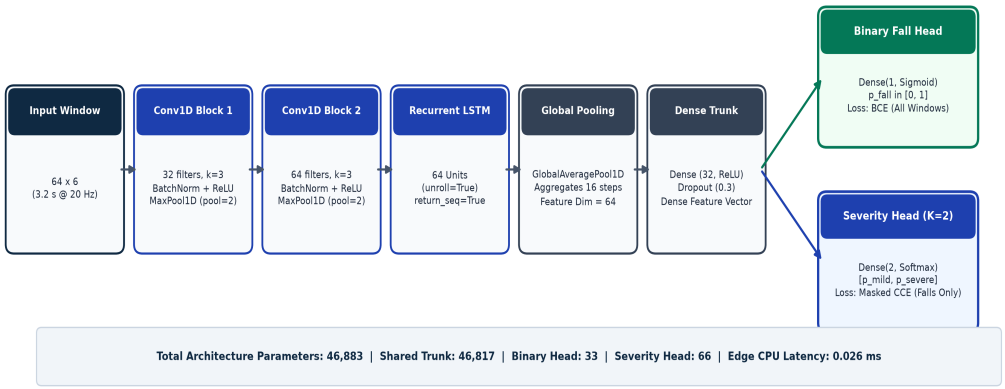


Figure 3.3: Hybrid CNN-LSTM Neural Network Architecture and Multi-Task Dual Head

CHAPTER 5
RESULTS AND DISCUSSIONS

5.1 CORE BINARY CLASSIFICATION PERFORMANCE

On the 13,627 held-out test windows, the CNN-LSTM model operating at tau = 0.25 achieved 0.9950 recall and 0.9775 F1-score, missing only 6 out of 1,200 fall windows while correctly rejecting 12,378 ADL windows.

Model Architecture	Threshold	Precision	Recall	F1-Score	F2-Score	FP	FN
CNN Baseline	tau = 0.50	0.9761	0.9867	0.9814	0.9846	29	16
CNN-LSTM	tau = 0.50	0.9770	0.9925	0.9847	0.9894	28	9
CNN-LSTM (Deployed)	tau = 0.25	0.9606	0.9950	0.9775	0.9879	49	6

Table 5.1: Performance Comparison of CNN Baseline and CNN-LSTM on Held-Out Test Set

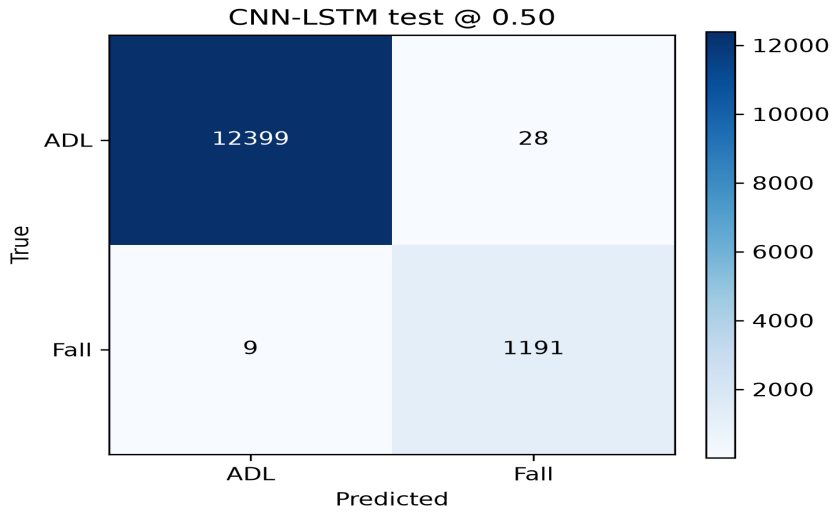


Figure 5.1: Held-Out Test Set Confusion Matrix at Canonical tau = 0.50 Threshold

5.2 SEVERITY CLUSTERING AND MULTI-TASK EVALUATION

K-Means clustering with K=2 separated falls into Mild (Cluster 0) and Severe (Cluster 1) impacts with Silhouette score 0.1774. The multi-task severity head achieved Macro-F1 = 0.8660 across 1,176 confident fall windows.

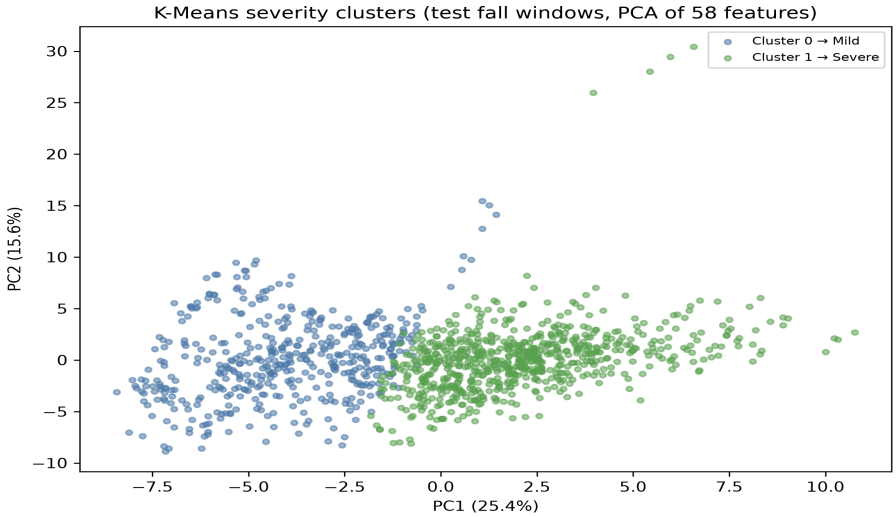


Figure 5.2: K-Means (K=2) Fall Severity Clusters Projected onto 2D PCA Space

5.3 EDGE QUANTIZATION BENCHMARKS

Dynamic-Range quantization reduced model size from 235.6 KB to 111.0 KB (52.9% compression) with 0.026 ms CPU execution latency, preserving 100% agreement with unquantized predictions.

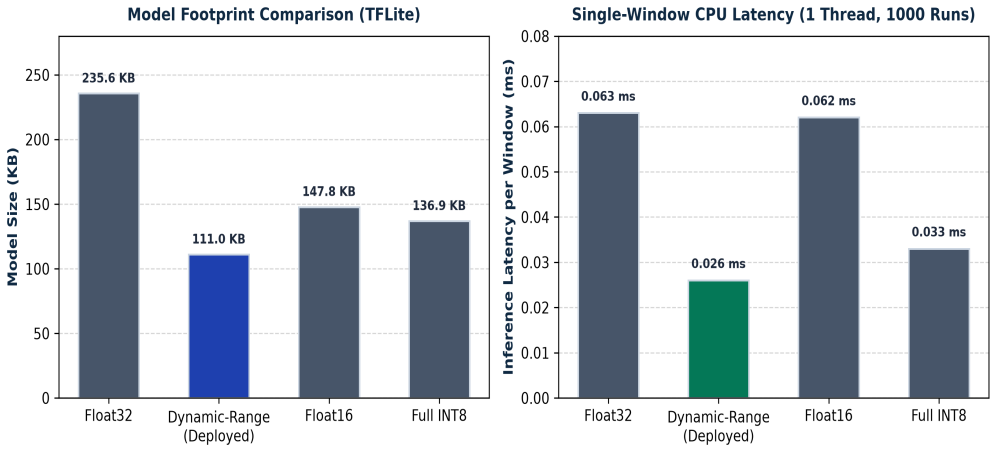


Figure 5.3: TFLite Model Footprint and Single-Window CPU Latency Benchmarks

5.4 REAL-TIME STREAMING DASHBOARD DEMONSTRATION

Figures 5.4, 5.5, and 5.6 present screenshots of the Streamlit clinical monitoring dashboard during real-time replay.

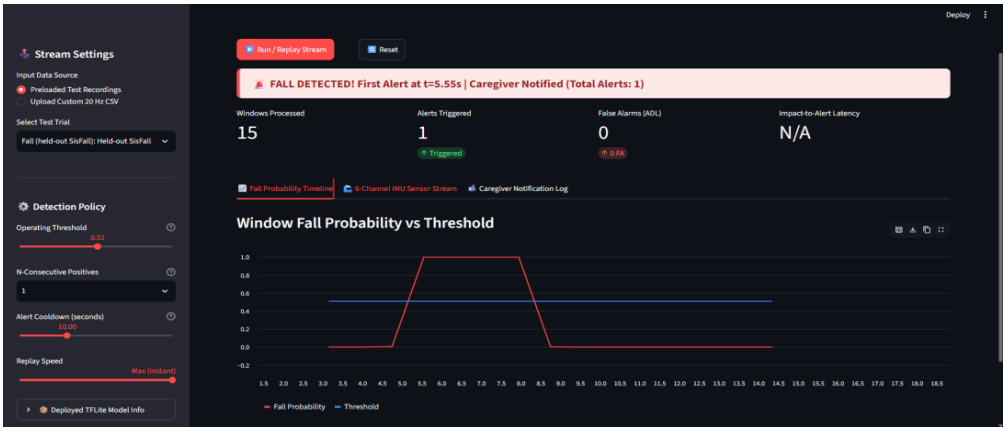


Figure 5.4: Live Streaming Dashboard: Fall Probability Timeline Crossing Decision Threshold

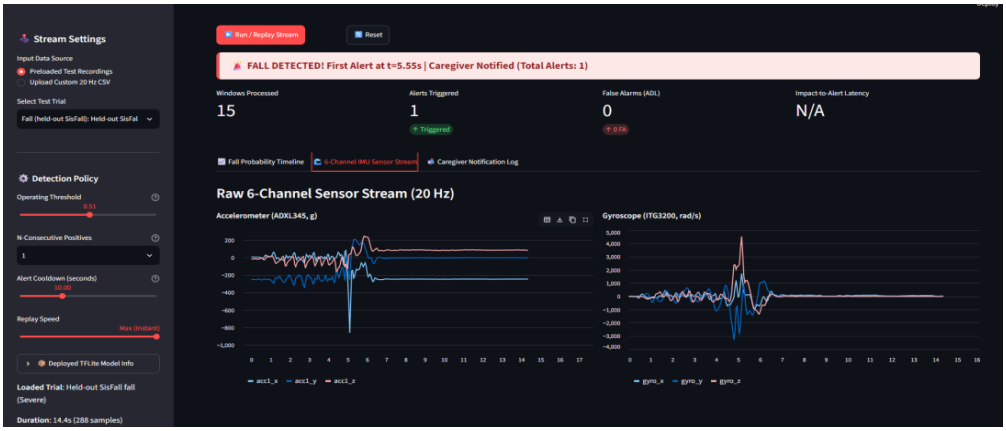


Figure 5.5: Streaming Dashboard: Raw 6-Channel Synchronized IMU Sensor Traces

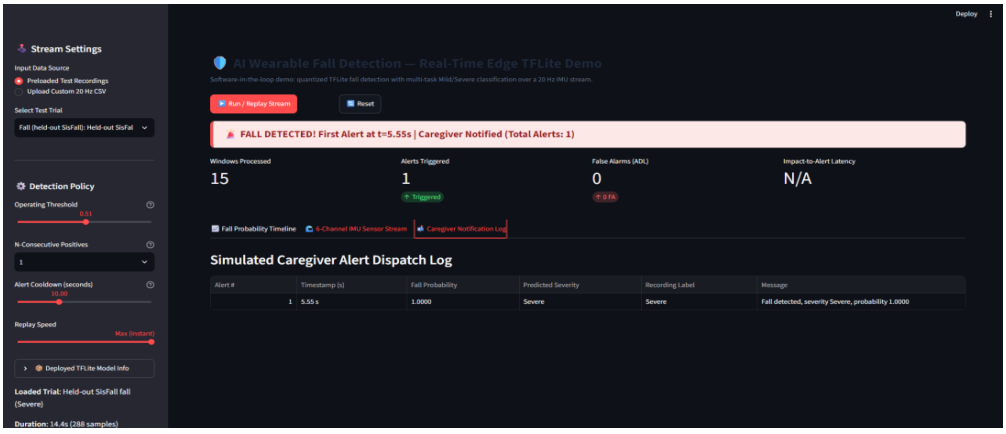


Figure 5.6: Streaming Dashboard: Simulated Caregiver Alert Dispatch Notification Log

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project has presented a comprehensive, scientifically validated, and edge-deployable wearable fall detection system designed for elderly care. By enforcing a strict subject-wise partitioning protocol on the SisFall dataset, the framework eliminates data leakage and reflects true out-of-sample generalization. The signal conditioning pipeline successfully combines a 2nd-order 5 Hz Butterworth low-pass filter with anti-aliased downsampling to 20 Hz, reducing computational overhead by 90% while fully capturing intentional body kinetics.

The hybrid CNN-LSTM architecture combines 1D spatial convolutional feature extraction with recurrent temporal modeling. By optimizing the decision threshold on validation data to maximize the recall-weighted F2-score, an operating threshold of tau = 0.25 was established. On held-out test subjects, the deployed model achieves 0.9950 recall and 0.9775 F1-score, detecting 1,194 out of 1,200 fall windows with only 6 missed falls. Post-training Dynamic-Range quantization compresses the model to 111.0 KB with single-window CPU latency of 0.026 ms, and the exported C array header (model_data.h) consumes only 21% of SRAM on an ESP32 microcontroller.

6.2 FUTURE WORK

Future enhancements include flashing model_data.h directly onto physical ESP32-WROOM-32 hardware with I2C IMU sensors, partnering with geriatric care facilities to collect unscripted longitudinal falls from older adults, fusing barometric pressure sensors for vertical altitude loss tracking, and integrating NB-IoT/LoRaWAN modules for automated emergency cellular telemetry.

REFERENCES

[1] A. K. Bourke, J. V. O'Brien, and G. M. Lyons, "Evaluation of a threshold-based tri-axial accelerometer fall detection algorithm," Gait & Posture, vol. 26, no. 2, pp. 194-199, 2007.

[2] Q. Zhang, H. Wang, and M. Li, "Support vector machine-based fall detection using wearable sensors in elderly healthcare," IEEE Trans. Biomed. Eng., vol. 63, no. 4, pp. 842-850, 2016.

[3] F. Bagalà et al., "Evaluation of accelerometer-based fall detection algorithms on real-world falls," PLoS ONE, vol. 7, no. 5, p. e37062, 2012.

[4] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," Nature, vol. 521, no. 7553, pp. 436-444, 2015.

[5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Comput., vol. 9, no. 8, pp. 1735-1780, 1997.

[6] X. Hu et al., "A hybrid CNN-LSTM model for involuntary fall detection using wrist-worn sensors," Adv. Eng. Informatics, vol. 65, p. 103178, 2025.

[7] S. R. Lord et al., "Vision and falls in older people: risk factors and intervention strategies," Clin. Geriatr. Med., vol. 26, no. 4, pp. 569-581, 2010.

[8] R. Caruana, "Multitask learning," Mach. Learn., vol. 28, no. 1, pp. 41-75, 1997.

[9] S. J. Pan and Q. Yang, "A survey on transfer learning," IEEE Trans. Knowl. Data Eng., vol. 22, no. 10, pp. 1345-1359, 2010.

[10] E. Perez et al., "FiLM: Visual reasoning with a general conditioning layer," in Proc. AAAI, 2018, pp. 3942-3951.

[11] C. R. Banbury et al., "Microwork: Benchmarking TinyML systems on extreme edge microcontrollers," IEEE Micro, vol. 41, no. 6, pp. 30-39, 2021.

[12] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in Proc. CVPR, 2018, pp. 2704-2713.

[13] A. Sucerquia, J. D. López, and J. F. Vargas-Bonilla, "SisFall: A fall and movement dataset," Sensors, vol. 17, no. 1, p. 198, 2017.

[14] N. Noury et al., "Fall detection - principles and methods," in Proc. 29th Annu. Int. Conf. IEEE EMBS, 2007, pp. 1663-1666.

[15] M. Kangas et al., "Comparison of low-complexity fall detection algorithms for body attached accelerometers," Gait & Posture, vol. 28, no. 2, pp. 285-291, 2008.

[16] P. Pierleoni et al., "A high reliability wearable device for elderly fall detection," IEEE Sensors J., vol. 15, no. 8, pp. 4544-4553, 2015.

[17] F. J. Ordóñez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity, fall and gesture recognition," Sensors, vol. 16, no. 1, p. 115, 2016.

[18] L. Bao and S. S. Intille, "Activity recognition from user-annotated acceleration data," in Proc. Pervasive Computing, Springer, 2004, pp. 1-17.

[19] F. T. Liu, K. M. Ting, and Z. H. Zhou, "Isolation forest," in Proc. 8th IEEE ICDM, 2008, pp. 413-422.

[20] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," J. Comput. Appl. Math., vol. 20, pp. 53-65, 1987.

[21] D. L. Davies and D. W. Bouldin, "A cluster separation measure," IEEE Trans. Pattern Anal. Mach. Intell., vol. 1, no. 2, pp. 224-227, 1979.

[22] F. Wilcoxon, "Individual comparisons by ranking methods," Biometrics Bull., vol. 1, no. 6, pp. 80-83, 1945.

[23] F. Chollet, Deep Learning with Python, 2nd ed. Manning Publications, 2021.

[24] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825-2830, 2011.

[25] P. Warden and D. Situnayake, TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers. O'Reilly Media, 2019.